

Ex 11.4

Q40 Area of parallelograms with vertices

$$\left. \begin{array}{l} A(1, 0, -1) \\ B(1, 7, 2) \end{array} \right\} \vec{AB} = \langle 0, 7, 3 \rangle$$

$$\left. \begin{array}{l} C(2, 4, -1) \\ D(0, 3, 2) \end{array} \right\} \vec{CD} = \langle -2, -1, 3 \rangle$$

These tell that $\vec{AB}$ and $\vec{CD}$ represent the diagonals

Geometric Interpretation:



$$\therefore \text{Area} = | \vec{AC} \times \vec{AD} |$$

$$= \left| \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 4 & 0 \\ -1 & 3 & 3 \end{vmatrix} \right|$$

$$= | 12 \underline{i} + (-3) \underline{j} + 7 \underline{k} |$$

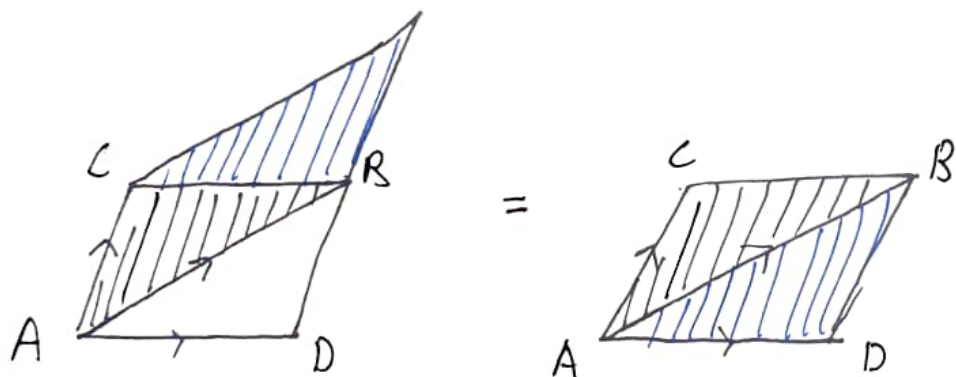
$$= \sqrt{12^2 + (-3)^2 + 7^2}$$

$$= \sqrt{202}$$

Remark I am unsure whether finding $\vec{AB}$ and $\vec{CD}$ and then sketch the diagram is needed.

If we use $|\vec{AC} \times \vec{AB}|$, we will get the same answer.

Rough idea why it works:



$$|\vec{AC} \times \vec{AB}| = |\vec{AC} \times \vec{AD}|$$